

Skills Summary

Arithmetic Sequences as Linear Functions

Use this reference while completing your worksheet or when studying.

Three questions, three tools. For an arithmetic sequence with first term a_1 and common difference d :

- What is term n ? $a_n = a_1 + (n - 1)d$
- Which term is this value? solve $a_1 + (n - 1)d = \text{value}$ for n
- What is the total of n terms? $S_n = \frac{n(a_1 + a_n)}{2}$

Read the question first: a term, a term *number*, and a total are three different answers.

1. The explicit formula for a term

$a_n = a_1 + (n - 1)d$. The $n - 1$ is the count of **steps**, not terms: reaching term 10 from term 1 takes nine steps. A decreasing sequence simply has a negative d — the formula never changes.

Example: Find the 10th term of the sequence with $a_1 = 6$ and $d = 4$.

Step 1. One specific term is wanted, so use the explicit formula and count steps rather than listing all ten terms.

Step 2. Nine steps separate term 1 from term 10, so $a_{10} = 6 + (10 - 1)(4) = 6 + 36$.

$$a_{10} = 42$$

Try it: Find the 8th term of the sequence with $a_1 = 11$ and $d = 5$.

check: 46

2. Solving for which term a value is

When the value is known and the position is not, set the explicit formula equal to the value and solve for n .

n must be a **positive whole number**. A fractional n means the value is not a term of the sequence at all — that is a real answer, not a mistake.

Example: Which term of 4, 10, 16, ... equals 58?

Step 1. The value is known but its position is not, so this is an equation to solve rather than a formula to evaluate.

Step 2. $4 + (n - 1)(6) = 58$ becomes $6n - 2 = 58$, so $6n = 60$.

$$n = 10$$

Try it: Which term of 9, 16, 23, ... equals 65?

check: 6

3. Summing a finite arithmetic sequence

$S_n = \frac{n(a_1 + a_n)}{2}$ — the number of terms times the **average** of the first and last.

The sum formula needs the *last* term, so compute a_n first whenever the problem gives only a_1 , d and n .

Example: Find the sum of the first 10 terms of 5, 8, 11, ...

Step 1. A total is wanted, so the sum formula applies — but it needs the last term, which is not given, so find that first.

Step 2. $a_{10} = 5 + 9(3) = 32$, and then $S_{10} = \frac{10(5 + 32)}{2} = 5 \times 37$.

$$S_{10} = \mathbf{185}$$

Try it: Find the sum of the first 12 terms of 2, 6, 10, ...

check: 882

4. A sequence as a linear function

Expanding $a_1 + (n - 1)d$ gives $f(n) = dn + (a_1 - d)$: a line with **slope** d , sampled only at $n = 1, 2, 3, \dots$

So the common difference is the slope, and the constant is the “term zero” value one step *before* the first term — not a_1 itself. That is the whole difference between discrete and continuous linear change.

Example: The sequence 7, 12, 17, ... is $f(n) = 5n + 2$. Find $f(12)$.

Step 1. The sequence is already in function form, so the term number is just an input — no stepping needed.

Step 2. Substituting the term number gives $f(12) = 5(12) + 2 = 60 + 2$.

$$f(12) = \mathbf{62}$$

Try it: The sequence 5, 11, 17, ... is $f(n) = 6n - 1$. Find $f(9)$.

check: 53

(!) Watch out: use $n - 1$, not n — one extra common difference is the most common wrong answer on this topic. Keep the sign of d from the story. And never multiply the number of terms by the *last* term to get a sum: multiply by the average of the first and last.